

Omar Montaser Youssef

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Education

University of East London & Ain Shams University - Double Degree Program

BSc in Computer and Artificial Intelligence Engineering, GPA: 3.4/4.0 (Excellent)

2023 – 2027

Relevant Courses: Distributed Computing, Network Operations and Management, Design of Operating Systems, Computer Networking, Internet of Things, Machine Learning, Advanced Algorithms and Complexity, Software Testing, Database Systems Design

Experience

Bayantech

Information Technology Intern

Oct – Nov 2025

- Worked with Google Cloud infrastructure and supported Sophos security administration in enterprise environments
- Managed virtual machines (Hyper-V, VMware) and improved RDP session handling for multi-user access
- Provided IT support (remote/on-site), resolving tickets while maintaining system configuration and asset tracking

Dar Al Fouad Hospital

Information Technology Intern

Aug – Oct 2025

- Restored hospital network during a major outage caused by corrupted VLAN configurations by rebuilding VLANs, reconfiguring switches, and resolving DHCP connectivity issues across six floors, ensuring uninterrupted SAP operations.
- Resolved IT tickets via TightVNC (OS installs, workstation recovery) and handled FortiGate firewall tasks
- Conducted asset inventory for 200 PCs, checking OS patch status, Kaspersky alerts, and system health

Certifications

CompTIA Security+ (SY0-701) 🔗

Google Cybersecurity Professional Certificate 🔗

AWS Cloud Solutions Architect Professional Certificate 🔗

Projects

Distributed LLM Inference System with Load Balancing & Fault Tolerance 🔗

- Architected a horizontally scalable inference system across heterogeneous compute nodes (GPU + CPU), exposed via Cloudflare Tunnels
- Deployed NVIDIA GPU workers running LLaMA 3.2 Instruct via vLLM, with Docker-based CPU fallback workers running DistilGPT-2
- Implemented a custom Weighted Least Response Time (WLRT) load balancer with connection pooling and per-node admission control, achieving 64 req/s throughput with 0% failure at high concurrency
- Built fault-tolerant scheduling with health checks, exponential backoff retries & dynamic request rerouting
- Deployed observability and monitoring stack (Prometheus, Grafana) with real-time system metrics

Secure Network Monitoring Lab (SNMPv3, Cisco, Wireshark) 🔗

- Built a Cisco-based network monitoring lab (GNS3, SnmpB, Wireshark) for SNMP management and traffic analysis
- Secured monitoring with SNMPv3 by configuring views, groups, users, and access control

IoT Home Intrusion Detection System (Cisco Packet Tracer) 🔗

- Designed an IoT security network in Packet Tracer using sensors, smart locks, and IP cameras.
- Developed 15+ automation rules to synchronize siren triggers and emergency access with sensor events.

Custom OS Kernel Memory Management (FOS) 🔗

- Implemented page fault handling and First Fit memory allocation in C via system calls
- Optimized virtual memory using working set management, page buffering, and Modified Clock replacement

Google Cybersecurity Projects 🔗

- Performed labs in vulnerability assessment, incident response, AAA, and NIST CSF application

Technical Skills

Systems & Programming: Linux, Windows, C, C++, Python, Java, SQL, Bash, Git

Cloud, Networking & Security: AWS, Docker, Redis, FastAPI, Django, Wireshark, Splunk, Burp Suite, Nmap, Suricata

Personal Accomplishments and Skills

- Hosted a Cloud Computing workshop on AWS at the Egyptian Chinese University
- Active in cybersecurity CTFs and TryHackMe labs, building hands-on exploitation and defense skills
- Achieved Specialist rank in codeforces, demonstrating strong problem-solving and algorithmic thinking